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Project Title:	Investigation of Interactions between Bioconvection and Natural Convection and Biofilm Growth in Porous Media		
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Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
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Comments:			
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Task Description:

This proposed research deals with one of the most vibrant areas of modern fluid dynamics, namely, microbiological fluid mechanics. Different from solid particles in traditional multiphase flow systems, many types of bacteria in bacterial suspension are self-propelled. They tend to swim in a particular direction in response to certain stimuli, such as gravity (gravitaxis), light (phototaxis), or chemical gradients (chemotaxis). Bacteria propel themselves by rotating flagella that are driven by reversible molecular motors that are embedded in the cell wall. In many cases the molecular motor periodically reverses its direction of rotation, producing on average irregular locomotion toward some specific direction. The number of these self-propelled bacteria in one cubic centimeter of suspension is usually very large (about E7 for a dilute regime up to about E11 for a close packed (turbulent) regime), which results in very complex flow patterns.
The proposed research project is divided into two parts. The first part deals with the development of a continuum model of bioconvection in a porous medium. This is a novel project because continuum model of bioconvection has been developed thus far only for microorganisms suspended in a clear fluid. Another novel feature of this project is the development of a model that describes the interaction between bioconvection and natural convection in a porous medium. The PI expects that this investigation will likely result in discovery of new instabilities, which do not occur when bioconvection only or natural convection only is studied. The developed model will include stochastic terms to account for the random element in bacterial motion. Accounting for randomness will shift the stability criteria, therefore the stability analysis will be repeated after stochastic terms are incorporated in the governing equations. The behavior and stability of the stochastic model will be investigated for different gravitational accelerations, including the microgravity case. It is expected that microgravity experiments will make it possible to validate the correctness of the stochastic terms in the equations, therefore results of this investigation will be used to plan a microgravity experiment.

The second part deals with direct simulation of bacterial motion in porous media and influence of bacterial motility and bioconvection on biofilm formation in porous rocks. This is also a novel project because nobody thus far tried to simulate directly bacterial swimming in three-dimensional configurations, as well as investigate such effects as collisions between two microorganisms and change in the direction of swimming of a bacterium when it enters a propulsive stream caused by another bacterium. The PI will also investigate interactions between bacteria and suspended solid particles. This means that he plans to investigate the stability of a suspension that includes two types of particles: (1) active, self-propelled microorganisms, and (2) passive solid particles. He will also investigate how bacterial and particle size distributions influence bacterial motion and biofilm formation.

Thus the proposed research is a very fundamental research program, which will result in obtaining new fundamental results in fluid physics. The objectives for this study are:

(1) Development of a continuum model of bioconvection in a porous medium. Investigation of the development of Rayleigh-Taylor instability

During this phase of the project, the PI will develop a model of bioconvection in porous media and analyze the development of Raleigh-Taylor plumes. Preliminary results obtained by the PI indicate that the development of this instability strongly depends on permeability of the porous medium (if permeability is very low, the instability does not develop). The PI plans to carry out a linear stability analysis of this problem and determine the critical value of permeability when the instability appears. He will also solve this problem numerically to investigate the development of gydrotactic plumes in bioconvection in a porous medium.

(2) Extending the continuum model to investigate the interaction between bioconvection and natural convection in a porous medium

At this stage, the model will be supplemented with an energy equation to investigate an interaction between bioconvection and natural convection. The PI expects that this investigation will result in discovery of new types of instabilities. Next, the model will be supplemented with the stochastic terms to account for the random element in bacterial motion. Accounting for randomness will shift the stability criteria, therefore the stability analysis will be repeated after stochastic terms are introduced in the equations. The behavior and stability of the stochastic model will be investigated for different gravitational accelerations, including the microgravity case. It is expected that microgravity experiments will make it possible to validate the correctness of the stochastic terms in the equations, therefore results of this investigation will be used to plan a microgravity experiment.

(3) Investigation of the effect of bioconvection on number density distribution of dead cells in a suspension of gyrotactic motile microorganisms that contains both living and dead cells.

This investigation is important to understand to what degree bioconvection complicates the separation between living and dead cells. If there were no bioconvection, living cells would concentrate in the upper part of the fluid layer and dead cells would concentrate in the lower part of the fluid layer. It should be noted that even if there were no convective motion of the fluid, the separation between the living and dead cells would not be perfect. This is because of the random component in the motion of microorganisms (which is modeled by cell diffusion). In the case of no bioconvection the distribution of the number density of living cells is determined by the balance between the upswimming of the cells and diffusion. For the same case, the distribution of the number density of dead cells is determined by the balance of buoyancy (that determines sinking velocity of the dead cells) and diffusion. For the case of no bioconvection the maximum number density of the living cells occurs at the free surface while the maximum number density of the dead cells occurs at the bottom of the fluid layer. If bioconvection is present, the living and dead cells are considerably more mixed up.

(4) Investigation of stability of suspensions of gyrotactic motile microorganisms in porous media and analyzing the effect of positioning porous plugs on suppressing bioconvection.

This task includes investigation of stability of a suspension of gyrotactic motile microorganisms in a fluid layer as well as the solution of the optimization problem of positioning porous plugs in a clear fluid domain to suppress bioconvection and optimize the separation between living and dead cells.

(5) Development of a model of biofilm formation in porous media

During this phase of the project, a model of biofilm accumulation will be developed. Initially, the model will be developed assuming that that all bacteria have a spherical shape, same radius, same density, and motility characteristics (such as the average velocity during the run intervals, and duration of the straight swimming and tumbling). Also, all solid particles will be assumed spherical having the same radius. Computations will be performed for both dilute regime when collisions between bacteria are negligible and also for the denser regime when some bacteria may collide or one

	bacterium may enter a propulsion flow of another bacterium. This research will result in gaining better understanding of how bioconvection interacts with natural convection, how random component of bacterial motion influences stability of bioconvection, how microgravity experiments can be used to validate the correctness of the stochastic terms in governing equations, how the motility of bacteria influences the dynamics of biofilm growth in a porous medium, how bioconvection influences the distribution of dead cells in suspensions of gyrotactic motile microorganisms that include both living and dead cells, and how porous blocks can be utilized to suppress bioconvection. This is an important fundamental investigation, which will lead to new physical knowledge concerning fluid motion induced by motile microorganisms and the stability of this motion. Investigation of bioconvection in composite porous/fluid domains will likely result in discovery of new instabilities, which do not occur in domains filled completely with a clear fluid or with a fluid saturated porous medium. The proposed investigation will generate new fundamental knowledge that will be useful for future applied research aimed at improving and controlling many biotechnological processes, including the microbial-enhanced oil recovery process.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research has a broad an impact on biotechnology. This is because this research suggests a method of mixing micro volumes of fluid. This will have a positive impact on pharmaceuticals, public health, laboratory testing, agriculture, and other areas where the ability to move, pump, mix, and otherwise manipulate micro volumes of fluids is important. In addition to this, this research further advances the area of microbiological fluid mechanics, which is one of the most vibrant areas of modern fluid dynamics.
Task Progress:	We finished the development of the concept of effective diffusivity in bioconvection. The results are presented in two of our journal papers (P. Geng and A.V. Kuznetsov, Introducing the Concept of Effective Diffusivity to Evaluate the Effect of Bioconvection on Small Solid Particles, International Journal of Transport Phenomena, vol. 7, pp. 321-338, 2005 and P. Geng and A.V Kuznetsov, Settling of Bidispersed Small Solid Particles in a Dilute Suspension Containing Gyrotactic Micro-Organisms, International Journal of Engineering Science, vol. 43, pp. 992-1010, 2005). We also analyzed the effect of vibration on stability of bioconvection. The results are presented in a journal paper (A.V. Kuznetsov, The Onset of Bioconvection in a Suspension of Negatively Geotactic Microorganisms with High-Frequency Vertical Vibration, International Communications in Heat and Mass Transfer, vol. 32, pp. 1119-1127, 2005) and were also presented at 2005 ASME International Congress and Exhibition (A.V. Kuznetsov, Investigation of Stability of a Suspension of Motile Microorganisms Subjected to a High-Frequency Vibration, Proceedings of 2005 International Mechanical Engineering Congress and Exhibition, Orlando, Florida, November 5-11, 2005, ASME paper IMECE2005-79904, pp. 1-5). A considerable effort was spent on investigating the interaction of bioconvection and natural convection. The results of this work are published in two journal papers (A.V. Kuznetsov, Investigation of the Onset of Thermo-Bioconvection in a Suspension of Oxytactic Microorganisms in a Shallow Fluid Layer Heated from Below, Theoretical and Computational Fluid Dynamics, vol. 19, pp. 287-299, 2005 and A.V. Kuznetsov, Thermo-Bioconvection in a Suspension of Oxytactic Bacteria, International Communications in Heat and Mass Transfer, vol. 32, pp. 991-999, 2005). We also investigated bioconvection in superimposed fluid and porous layers. The results are presented in the following journal paper (A.A. Avramenko and A.V. Kuznetsov, Linear Instability Analysis of a Suspension of Oxytactic Bacteria in Superimposed Fluid and Porous Layers, Transport in Porous Media, vol. 61, pp. 157-175, 2005). Finally, our investigations of stability of suspensions of gyrotactic microorganisms are published in the following papers (A.V. Kuznetsov, The Onset of Bioconvection in a Suspension of Gyrotactic Microorganisms in a Fluid Layer of Finite Depth Heated From Below, International Communications in Heat and Mass Transfer, vol. 32, pp. 574-582, 2005; A.V. Kuznetsov and P. Geng, The Interaction of Bioconvection Caused by Gyrotactic Micro-Organisms and Settling of Small Solid Particles, International Journal of Numerical Methods in Heat and Fluid Flow, vol. 15, pp. 328-347, 2005; and A.V. Kuznetsov and A.A. Avramenko, Effect of Fouling on Stability of Bioconvection of Gyrotactic Microorganisms in a Porous Medium, Journal of Porous Media, vol. 8, pp. 45-53, 2005).
Bibliography Type:	Description: (Last Updated: 04/27/2010)
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